



ANALYSTLAB AFRICA

Data Science Internship Programme

Official Weekly Assignment

Week 2: Feature Engineering & Data Preprocessing for Machine Learning

Document Control

Programme	Data Science Internship
Week	Week 2
Project	Feature Engineering & Data Preprocessing for Machine Learning
Version	1.0
Issue Date	_____
Submission Deadline	End of Week 2 (Sunday, 11:59 PM WAT)

Business Scenario

You are working as a Junior Data Scientist at AnalystLab Africa Consulting.

Following your successful exploratory data analysis in Week 1, your project manager has requested that you prepare the dataset for machine learning.

Before a machine learning model can be trained, datasets must be cleaned further, transformed, encoded, scaled, and engineered into a format suitable for predictive modelling.

Your objective this week is to transform raw business data into a machine-learning-ready dataset while documenting every preprocessing decision.

Learning Objectives

By the end of this assignment, you should be able to:

- Understand the importance of feature engineering.
- Identify useful and irrelevant features.
- Handle categorical and numerical variables.
- Encode categorical variables appropriately.
- Scale and normalize numerical features.
- Detect and handle outliers.
- Perform feature selection.
- Prepare datasets for machine learning.

Dataset

Choose **ONE** of the following datasets.

Option 1

Heart Disease Prediction Dataset

<https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction>

Objective

Prepare the dataset for predicting whether a patient is likely to develop heart disease.

Option 2

Loan Prediction Dataset

<https://www.kaggle.com/datasets/altruistdelhite04/loan-prediction-problem-dataset>

Objective

Prepare the dataset for predicting whether a loan application should be approved.

Business Questions

Your analysis should answer the following questions:

1. Which features are most relevant to the prediction problem?
2. Which variables require encoding?
3. Which variables require scaling or normalization?
4. Are there any redundant or highly correlated features?
5. How should missing values and outliers be handled?
6. What preprocessing techniques improve the dataset quality?
7. Is the dataset ready for machine learning?

Assignment Tasks

Part 1 – Business Understanding

Research the prediction problem associated with your chosen dataset.

Prepare a **1–2 page Business Understanding Report** explaining:

- Business problem
- Project objective
- Target variable

- Importance of predictive modelling
- Expected business impact

Part 2 – Data Inspection

Using Python:

- Load the dataset.
- Display the first 10 rows.
- Examine the dataset structure.
- Identify:
 - Rows and columns
 - Data types
 - Missing values
 - Duplicate records
 - Summary statistics

Part 3 – Feature Engineering & Data Preprocessing

Prepare the dataset by performing:

Data Cleaning

- Handle missing values.
- Remove duplicates.
- Correct data types.

Feature Engineering

- Create new features where appropriate.
- Remove irrelevant features.
- Rename columns for clarity.

Feature Encoding

Apply suitable encoding techniques such as:

- Label Encoding
- One-Hot Encoding

Explain why you selected each technique.

Feature Scaling

Apply one or more of:

- StandardScaler
- MinMaxScaler
- RobustScaler

Explain why your chosen scaling method is appropriate.

Outlier Detection

Detect and handle outliers using techniques such as:

- Boxplots
- IQR Method
- Z-score

Explain your approach.

Part 4 – Feature Selection

Identify:

- Most important features.
- Highly correlated variables.
- Features that should be removed.

Support your decisions with:

- Correlation Heatmap
- Feature Importance Analysis (if applicable)

Part 5 – Final Machine Learning Dataset

Produce a final dataset that is ready for machine learning.

Clearly explain every preprocessing step applied

Minimum Project Requirements

To ensure consistency and maintain a professional standard across all submissions, your project **must** meet the following minimum requirements:

Python Notebook

Your Jupyter Notebook must include:

- Clear section headings using Markdown.
- Well-documented Python code with meaningful comments.
- Outputs displayed for each significant step.
- Logical workflow from data loading to final preprocessing.

Data Preprocessing

Your preprocessing pipeline must include:

- Missing value assessment and appropriate handling.
- Duplicate record detection and removal (where applicable).
- Data type validation and correction.
- Categorical variable encoding.
- Feature scaling or normalization.
- Outlier detection and treatment.
- Feature selection or feature engineering (where appropriate).

Visualizations

Include **at least 6 visualizations**, such as:

- Histograms
- Box Plots
- Correlation Heatmap
- Count Plots
- Scatter Plots
- Pair Plots (optional)

All visualizations should have:

- Clear titles

- Proper axis labels
- Appropriate legends (where applicable)

Reports

Your submission must include:

- Business Understanding Report
- Data Preprocessing Report

Each report should clearly explain:

- The business problem.
- Preprocessing decisions.
- Feature engineering techniques.
- Key observations.
- Justification for the methods used.

Project Organization

Your GitHub repository must include:

- `README.md`
- Jupyter Notebook (`.ipynb`)
- Original dataset (if permitted)
- Cleaned dataset (`.csv`)
- Machine-learning-ready dataset (`.csv`)
- Reports
- Supporting images (if used)

Tools to Use

Required Tools

- Python
- Jupyter Notebook or Google Colab

Libraries

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

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Deliverables

Submit the following:

1. Cleaned Dataset (.csv)
2. Machine Learning Ready Dataset (.csv)
3. Jupyter Notebook (.ipynb)
4. Business Understanding Report (PDF or Word)
5. Data Preprocessing Report (PDF or Word)
6. GitHub Repository
7. Upload all files to Google Drive (Anyone with the link can view)
8. Submit using the official Week 2 Submission Form

• Submit using the official Week 2 Submission Form Assessment Criteria

Criteria	Weight
Business Understanding	10%
Dataset Inspection	15%
Feature Engineering	25%
Data Preprocessing	20%

Feature Selection	15%
Professionalism & Documentation	15%

Professional Development Requirement

As part of building your professional portfolio:

- Update your GitHub repository with your Week 2 project.
- Publish one LinkedIn post describing:
 - The dataset used
 - Feature engineering techniques applied
 - Key preprocessing decisions
 - Skills developed
- Tag **AnalystLab Africa** on LinkedIn.
- Publish one X (Twitter) post summarizing your project.
- Tag **@analystlabafri**c on X.
- Include the hashtag **#AnalystLabAfrica** on both LinkedIn and X.

Professional Branding Reminder

AnalystLab Africa encourages every intern to document their learning journey professionally on LinkedIn and X. Consistently sharing your work helps build your portfolio and increases your visibility to recruiters and employers.

Submission Checklist

- ☐ Business Understanding Report
- ☐ Data Preprocessing Report
- ☐ Cleaned Dataset (.csv)
- ☐ Machine Learning Ready Dataset (.csv)
- ☐ Jupyter Notebook (.ipynb)
- ☐ GitHub Repository Updated
- ☐ LinkedIn Post Published
- ☐ X Post Published
- ☐ Google Drive Link Submitted